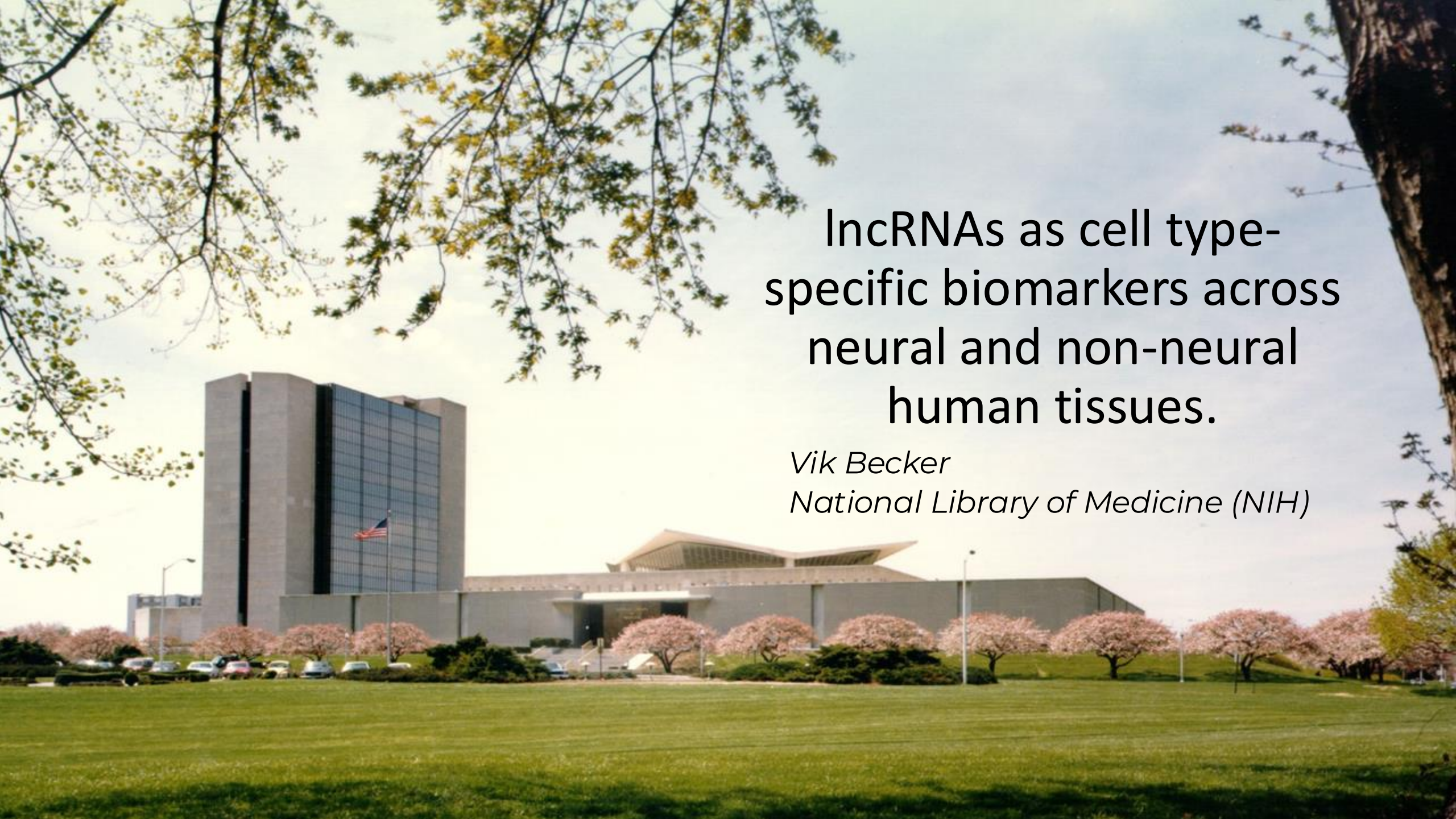


A photograph of the National Library of Medicine (NIH) building, a large, modern, multi-story structure with a prominent glass facade. The building is surrounded by a green lawn and several trees, including some with pink blossoms. The sky is blue with some clouds. The text "IncRNAs as cell type-specific biomarkers across neural and non-neural human tissues." is overlaid on the right side of the image.

IncRNAs as cell type-specific biomarkers across neural and non-neural human tissues.

Vik Becker

National Library of Medicine (NIH)

Overview

- I. Introduction to lncRNA and NS-Forest
- II. Data and Methods
- III. Neural vs Non-Neural Findings
- IV. Retina-Specific Findings
- V. Conclusions and Future Directions

I. Introduction to lncRNA and NS-Forest

Non-Coding RNA (ncRNA)

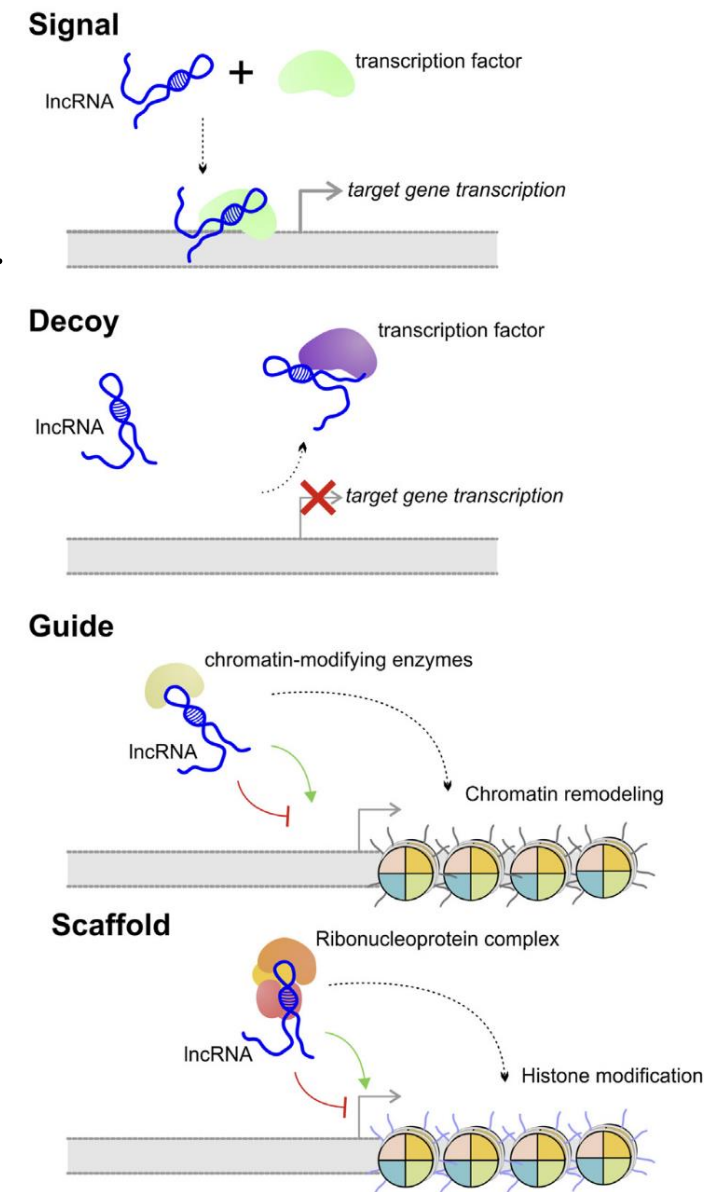
= transcriptional products that do not encode proteins.

- Though it's estimated that **~75%** of the human genome is transcribed, only **~2%** is protein coding!
- ncRNA *initially* considered an RNA processing byproduct.
- Regulatory ncRNA Discovery Timeline:
 - **1990**: First human regulatory lncRNA, *H19*, identified.
 - **2007**: Encyclopedia of DNA Elements (ENCODE) project initiation.
 - **2019**: First FDA-approved diagnostic lncRNA biomarker.

Long Non-Coding RNA

= non-coding RNA transcripts greater than 200 nucleotides long.

- Generally abbreviated to **lncRNA**.
- **Regulate** gene expression through diverse nuclear and cytoplasmic mechanisms.
- lncRNA genes generally exhibit **lower expression** but **greater cell-type specificity**.
 - **Specificity** is a key trait for biomarker candidates.
 - Many lncRNA genes only annotated recently.
- Utility for distinguishing healthy cell types remains **poorly characterized**.



NS-Forest v4.1

= identifies minimal marker gene sets (biomarkers) that classify cell types using random forest models.



- Candidate biomarkers prioritized based on **positive binary expression patterns**.
 - Binary Score = **[0,1]** measure of the specificity of a gene's expression to a given cell type.
 - **0** = gene is *more highly expressed in every other cell type* in comparison to the target.
 - **1** = gene is *only* expressed in the target cell type
- In NS-Forest runs on diverse tissues, it frequently identified **lncRNAs** as cell type specific markers *if the tissue was enriched in neuronal* cell types.

We aimed to evaluate:

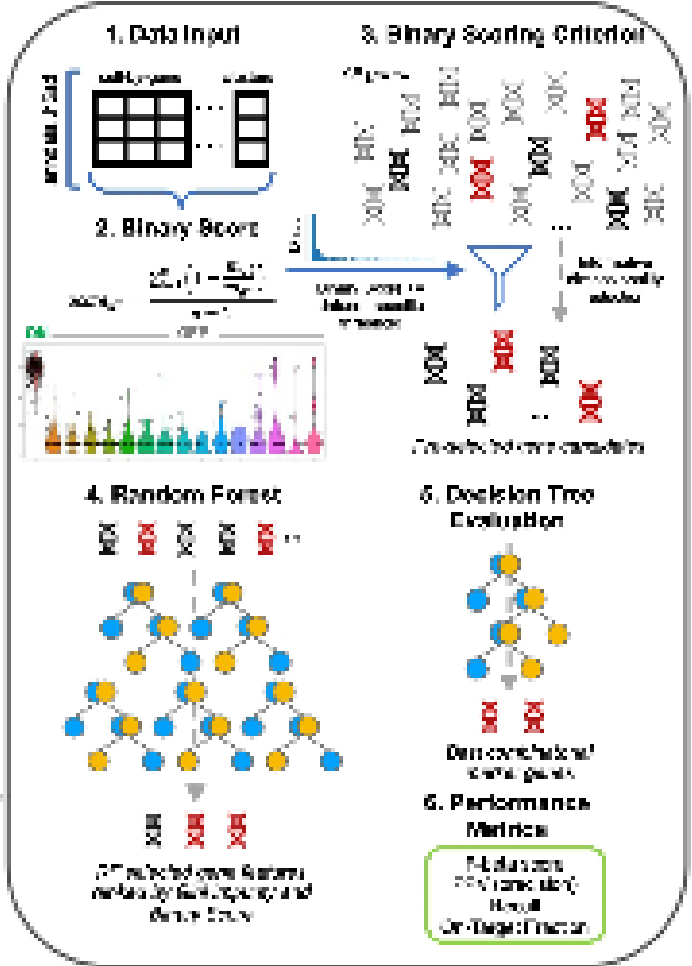
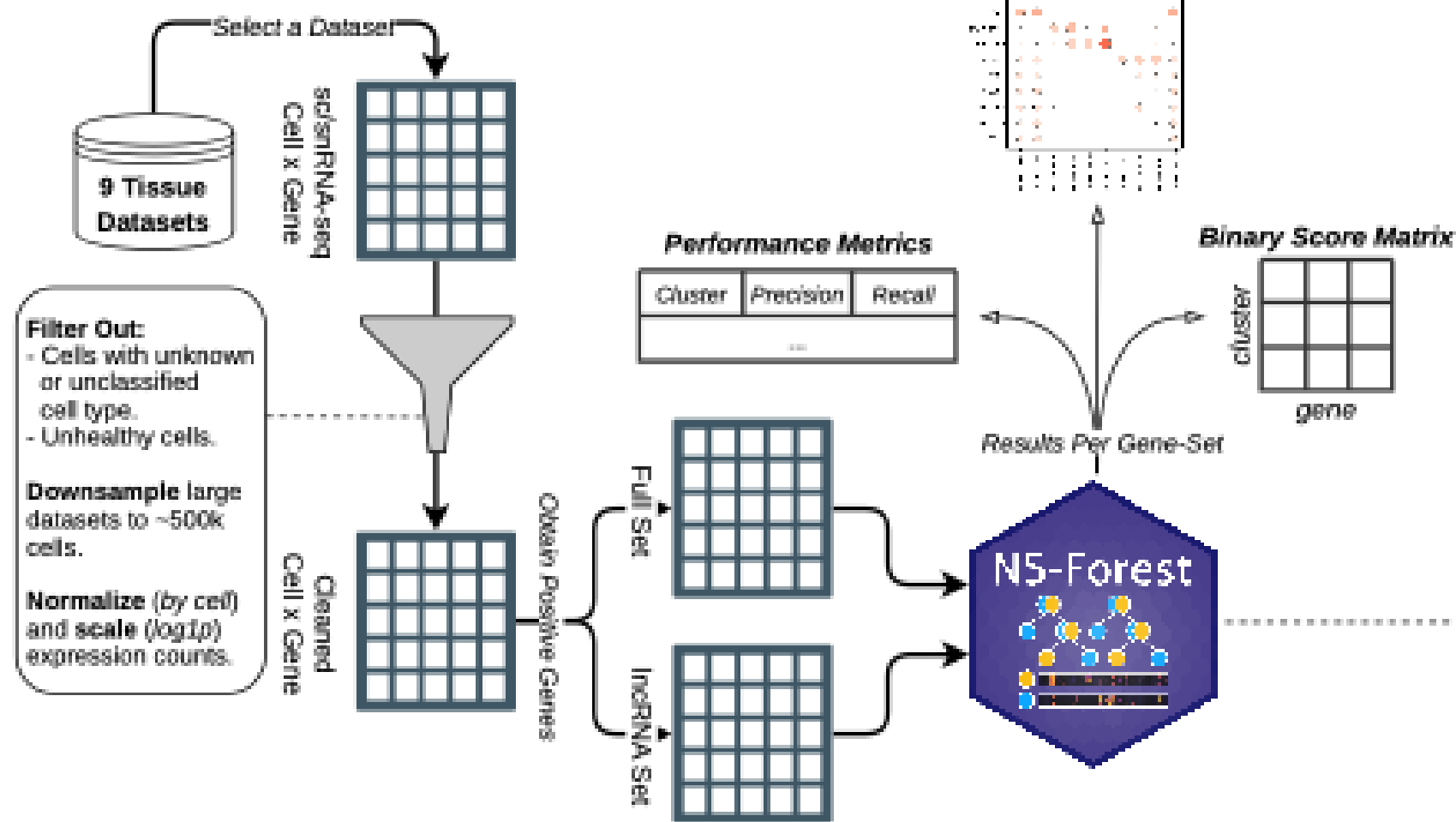
- Whether lncRNAs alone retain sufficient **discriminatory information** to **classify** cell types.
- Whether lncRNA **biomarker potential**...
 - ...differs between ***neural and non-neural*** healthy human tissues.
 - ...is particularly pronounced for any specific ***retinal cell types***.

II. Data and Methods

Descriptive Table of Cell x Gene Tissue Datasets

<i>Type</i>	<i>Tissue</i>	<i>Number of Cell Types</i>	<i>Number of Cells (cleaned)</i>	<i>Gene Set</i>	<i>Number of Genes</i>
Non-Neural	Bone Marrow	45	66.6k	Full	49,082
				lncRNA	12,128
	Breast	42	507.0k	Full	33,604
				lncRNA	14,399
	Kidney	75	107.7k	Full	33,140
				lncRNA	11,509
	Liver	29	259.6k	Full	35,659
				lncRNA	11,560
	Lung	61	491.2k	Full	27,402
				lncRNA	8,744
Neural	Motor Cortex	127	76.5k	Full	30,751
				lncRNA	2,032
	Middle Temporal Gyrus	75	15.6k	Full	30,751
				lncRNA	2,032
	Retina	123	509.0k	Full	35,475
				lncRNA	15,491
	Spinal Cord	43	28.6k	Full	35,467
				lncRNA	15,488

NS-Forest Pipeline



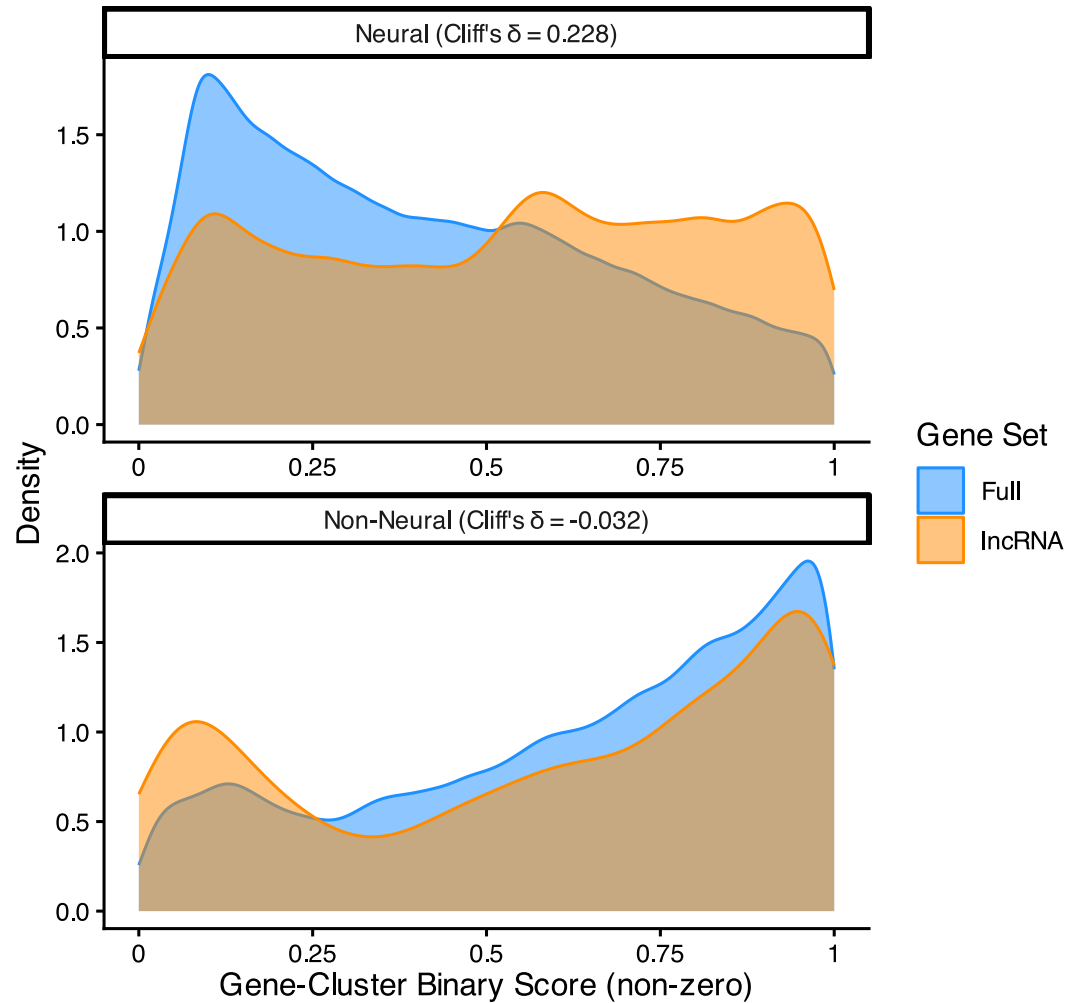
III. Neural *versus* Non-Neural Findings

Neural tissues, in comparison to **non-neural**...

- Contain a **larger proportion of positive lncRNAs**.
 - lncRNAs have a **relatively larger contribution** to the pool of candidate neural biomarkers and **higher median expression** in neural cell types.
- Retain a **larger** average number of **candidate lncRNA biomarkers** per cell type.
 - Indicates **more diversity** in cell type-specific lncRNA expression.

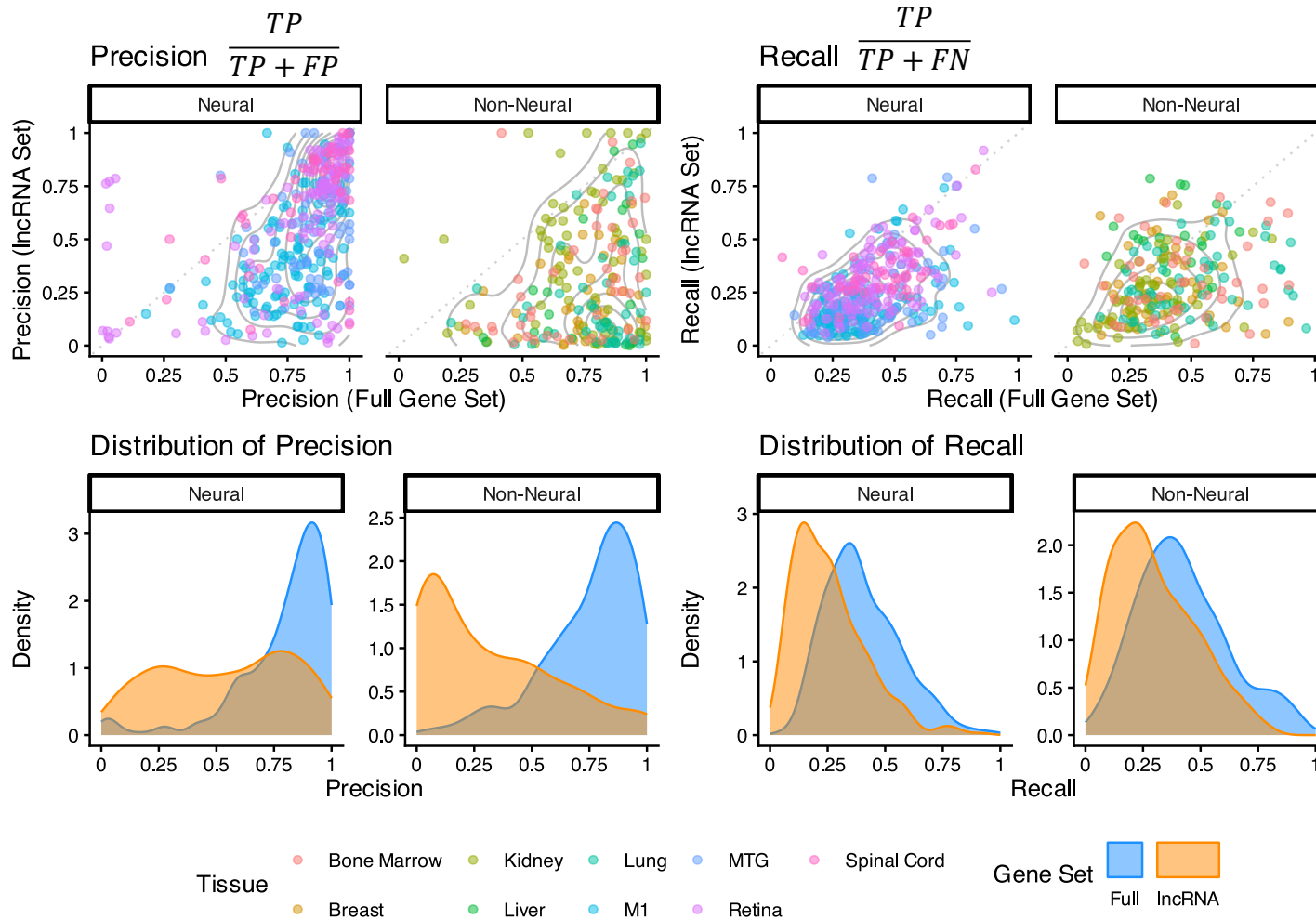
Type	Tissue	Gene Set	Number of Positive Genes (% of Total)	Mean Number of Candidate Genes / Cell Type
Non-Neural	Bone Marrow	Full	6,474 (13.2%)	622.1
		lncRNA	79 (0.7%)	7.3
	Breast	Full	4,622 (13.8%)	425.6
		lncRNA	64 (0.4%)	5.1
	Kidney	Full	5,249 (15.8%)	465.6
		lncRNA	203 (1.8%)	12.8
	Liver	Full	3,073 (8.6%)	295.4
		lncRNA	41 (0.4%)	3.2
	Lung	Full	5,288 (19.3%)	461.9
		lncRNA	83 (1.0%)	4.8
Neural	Motor Cortex	Full	8,723 (28.4%)	651.5
		lncRNA	262 (12.9%)	22.0
	Middle Temporal Gyrus	Full	11,529 (37.5%)	750.9
		lncRNA	386 (19.0%)	32.7
	Retina	Full	6,925 (19.5%)	590.6
		lncRNA	703 (4.5%)	56.5
	Spinal Cord	Full	8,425 (23.8%)	515.2
		lncRNA	703 (4.5%)	61.6

Distribution of Non-Zero Binary Scores



- **Neural** cells show **moderate separation** ($\delta = 0.228$; *lncRNA subset is higher*).
 - lncRNAs show a **greater proportion** of high Binary Scores compared to the full transcriptome, consistent with **greater cell type-specific expression**.
- **Non-neural** cells show **negligible separation** ($\delta = -0.032$; *full transcriptome set is higher*)
 - The Binary Score distributions are **broadly comparable**, suggesting that the two gene sets are **similarly specific**.

NS-Forest Performance Metrics by Gene Set.



- Restricting the gene set to lncRNAs **overall reduces precision**, indicating an expected **loss of information** when removing protein-coding genes.
- Better precision preservation** in **neural** tissues indicates a greater retention of **discriminatory information** in the lncRNA gene subset.

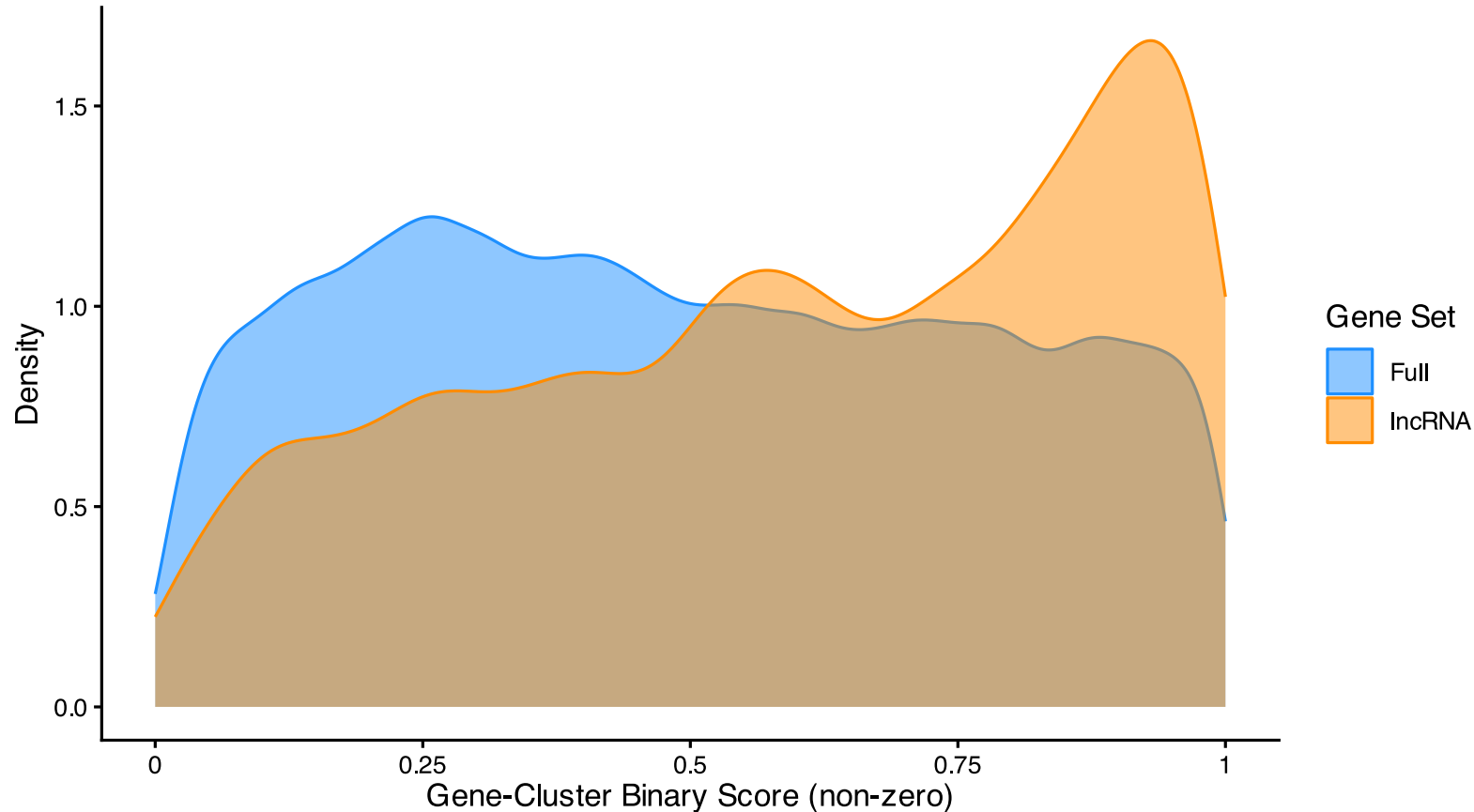
IV. Retina-Specific Findings

Of the positive genes of each tissue, the **proportion** of positive lncRNAs is **larger in retina** than any other tissue.

- **Increased contribution** of lncRNAs to the retina candidate biomarkers pool.
- Indicates comparably **higher non-zero lncRNA expression**.

Type	Tissue	Gene Set	Number of Positive Genes	% of Positive Genes that are lncRNA	Mean Number of Candidate Genes / Cell Type
Non-Neural	Bone Marrow	Full	6,474 (13.2%)	1.22%	622.1
		lncRNA	79 (0.7%)		7.3
	Breast	Full	4,622 (13.8%)	1.38%	425.6
		lncRNA	64 (0.4%)		5.1
	Kidney	Full	5,249 (15.8%)	3.87%	465.6
		lncRNA	203 (1.8%)		12.8
	Liver	Full	3,073 (8.6%)	1.33%	295.4
		lncRNA	41 (0.4%)		3.2
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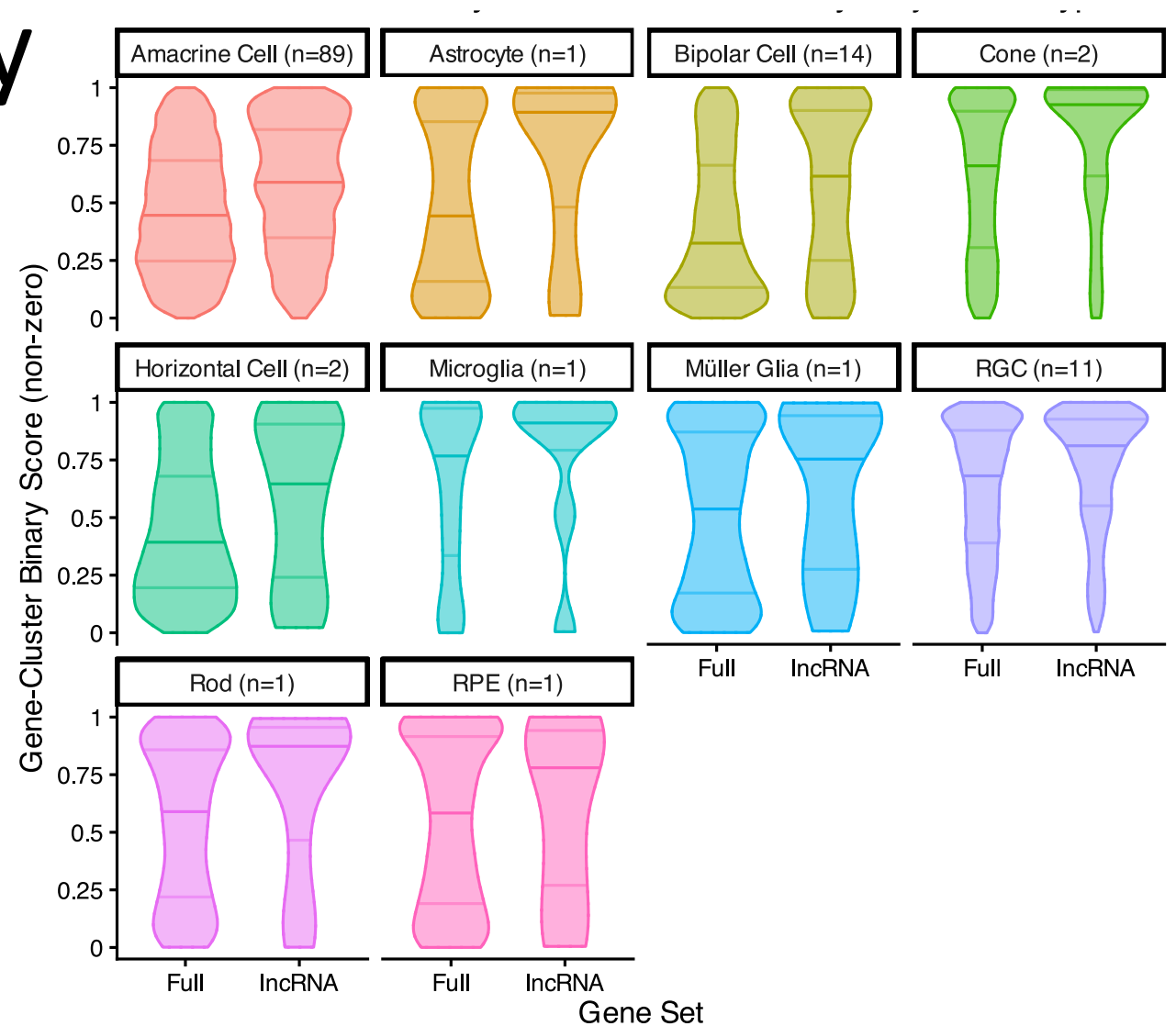
Distribution of Non-Zero Retina Binary Scores



- Distributions have a Cliff's $\delta = 0.224$ (*moderate separation; lncRNA subset is higher*).
 - lncRNAs show a **greater proportion** of high Binary Scores compared to the full transcriptome.
 - Indicates that retinal lncRNA having **greater cell type-specific expression**.

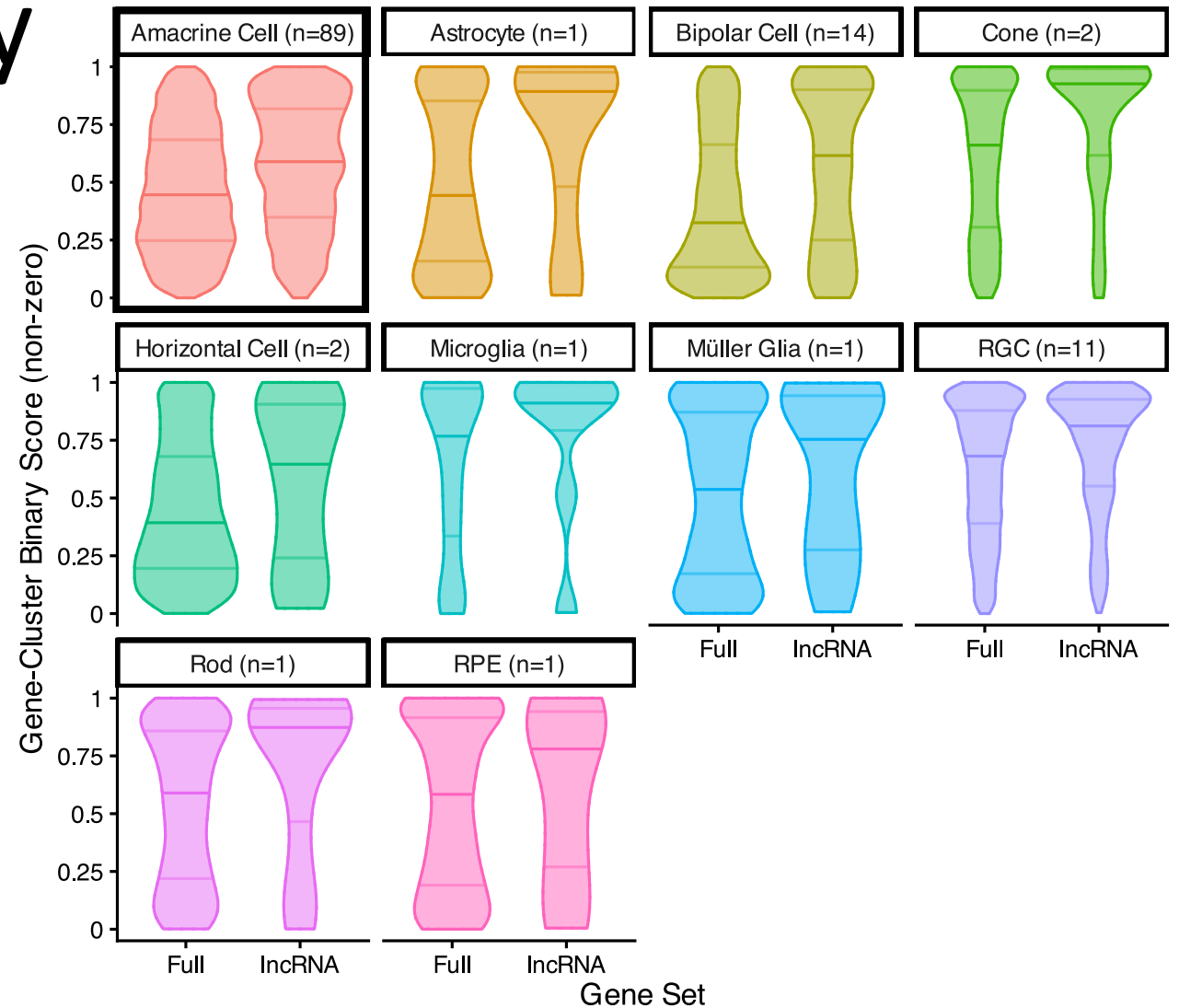
Retina Non-Zero Binary Score Distributions by Major Cell Type Class

- The **median** Binary Score is **higher in the lncRNA set** across **all** major cell type classes.
 - Indicates that lncRNA **cell-type specificity** is generally **high** in retinal cells, particularly in those with more differential central tendencies.
 - *EX: Astrocytes, Bipolar Cell, Rod.*

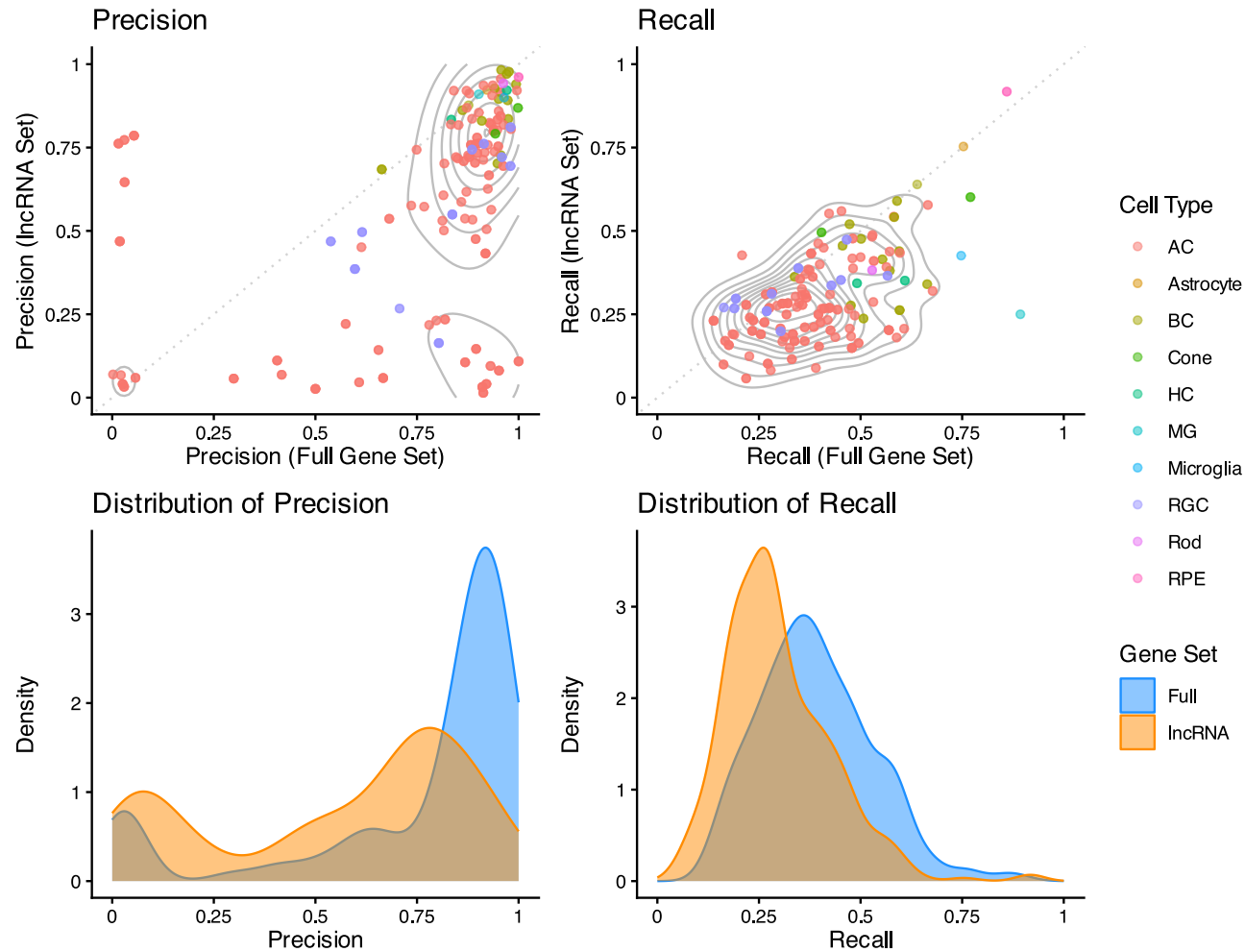


Retina Non-Zero Binary Score Distributions by Major Cell Type Class

- The more **uniform** non-zero Binary Score distribution of **amacrine cells** may reflect the large number of **closely-related subtypes** within the class.
 - lncRNAs **shared** across multiple cell types exhibit **lower Binary Scores** because their expression is distributed across the cell types.



NS-Forest Performance Metrics



- Restricting the gene set to lncRNAs **generally reduces precision and recall.**
- Precision appears **largely preserved** in the lncRNA set.
- *Key Class Observations:*
 - High diversity in retention among **Amacrine** cell types.
 - The lncRNA set retains high precision and recall for **RPE** and **Astrocyte** clusters.

Top 10 Cell-Type Biomarkers in the lncRNA Set by On-Target Fraction

$$\text{OnTarget Fraction}_{gT} = \frac{\text{median expression in target cluster}}{\text{sum of median expression across all clusters}} = \frac{m_{gT}}{\sum_{i=1}^n m_{gi}}$$

Cell Type	Gene Symbol	On-Target Fraction
RPE	LINC00276	0.952
Astrocyte	ENSG00000243620	0.883
OFFx	FEZF1-AS1	0.860
OFFx	ENSG00000278254	0.860
DB3a	ENSG00000267109	0.842
DB3a	LINC01091	0.842
BB	ENSG00000250971	0.816
S_Cone	RAB17-DT	0.798
S_Cone	ENSG00000250421	0.798
VIP	ENSG00000286481	0.782

- **LINC00276** (1) and **FEZF1-AS1** (3) have published functions in the retina.
 - **LINC00276**: key lncRNA in age-related macular degeneration (AMD).
 - **FEZF1-AS1**: oncogenic lncRNA in retinoblastoma.
- **ENSG00000243620** (2) is associated with key neural development transcription factors *ZIC1*, *ZIC2*, and *ZIC4*.
- No other published retinal/neural associations.

V. Conclusions and Future Directions

Conclusions: *Neural vs Non-Neural*

- **Neural tissues** contain **more candidate marker genes** than **non-neural tissues**, with a larger contribution from lncRNAs, suggesting that **lncRNAs** are **more highly expressed in neural-related cells**.
- **lncRNAs** exhibit **greater cell type specificity** than the full transcriptome in **neural tissues**.
- Restricting analyses to lncRNAs **generally decreases** the **precision** of NS-Forest classification, but the **reduction is smaller** in **neural** tissues than in **non-neural** tissues.
- **lncRNAs** may play a unique role in establishing the **highly specific gene regulation** necessary to **generate** the large number of distinct **neural** cell subtypes.

Conclusions: *Retina-Specific*

- AS

Future Directions

Comparison with downsampled coding genes (EX: do 79 coding genes retain the same discriminatory information as 79 lncRNA genes).

Collaboration with NEI on lncRNA retina biomarkers.

idk man 1-2 more bullet points here

Thank you!

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				lncRNA	15,491	703 (4.5%)		56.5
	Spinal Cord	43	28.6k	Full	35,467	8,425 (23.8%)	8.34%	515.2
				lncRNA	15,488	703 (4.5%)		61.6

